Video Anomaly Detection in Assembly/Production Lines

CSE 496

Second Presentation

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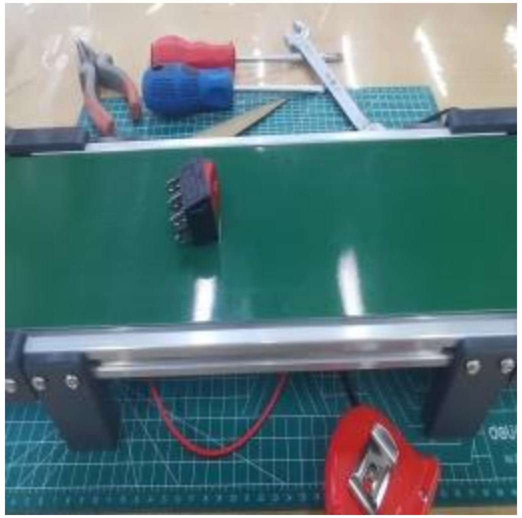
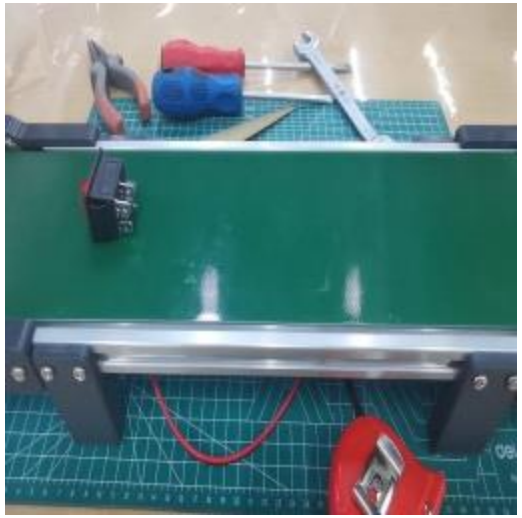
Project Definition

NORMAL



ANOMALY



TEXT Descriptions of anomaly using
Chain of Thought

- Detect anomalies on a production line.
- Replace manual visual inspection with an AI-based system that "watches" the line and flags anything wrong.
- The system learns what "normal" looks like through plain text instructions, so it can adapt to any product without retraining.

Approach: Zero-Shot VLMs Instead of Custom Training

Why a VLM?

- No labeled training set required
- Natural-language defect specs
- Adapts as the line spec evolves
- Produces interpretable rationales

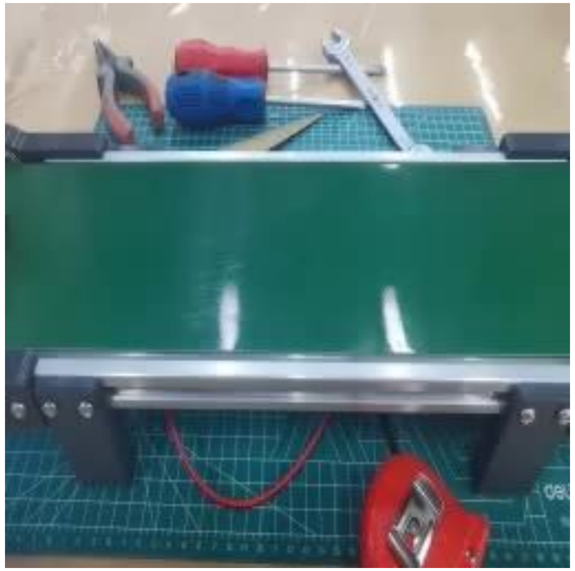
Why Open-Weight?

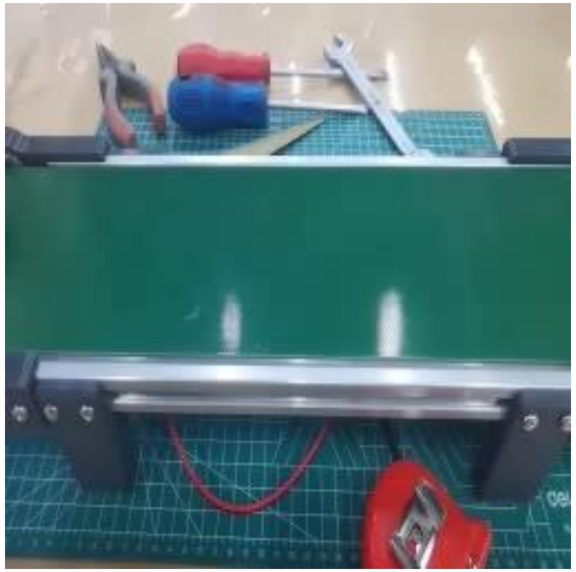
- Runs on-premise (no API egress)
- Reproducible (fixed weights)
- Cost: free after one-time GPU
- Data privacy preserved

Why Qwen VL Family?

- Native video input support
- Strong spatial reasoning
- Sizes 2B → 235B available

VAD Using QWEN VLM models





- Used a structured, step-by-step prompt.
- Tried with Qwen3.6-Plus with dashscope API.

Shorter Prompt Example:

```
VIDEO_PROMPT = """
You are a quality control vision system analyzing a conveyor-belt video.

Task:
Track one push-button across the full video and give one final verdict.

Reference:
- Red cap side = round red top cap
- Pin side = black base with 4 metal pins
- Normal orientation: red cap on the RIGHT, pin side on the LEFT
- Anomaly: reversed orientation, stopped, fell, or left the belt

Instructions:
1. Inspect all frames before deciding.
2. Check orientation, motion, and belt position.
3. Ignore partial visibility, slight wobble, and small off-center placement.

Answer format:
VERDICT: ✅ NO ANOMALY DETECTED / ❌ ANOMALY DETECTED
ANOMALY REPORT: [type, description, position, when, cause, severity or None]
"""
```

Different Types of Personas

- We have different personas inspecting the production line.
- They all have common entrance for Anomaly detection but roles are different.

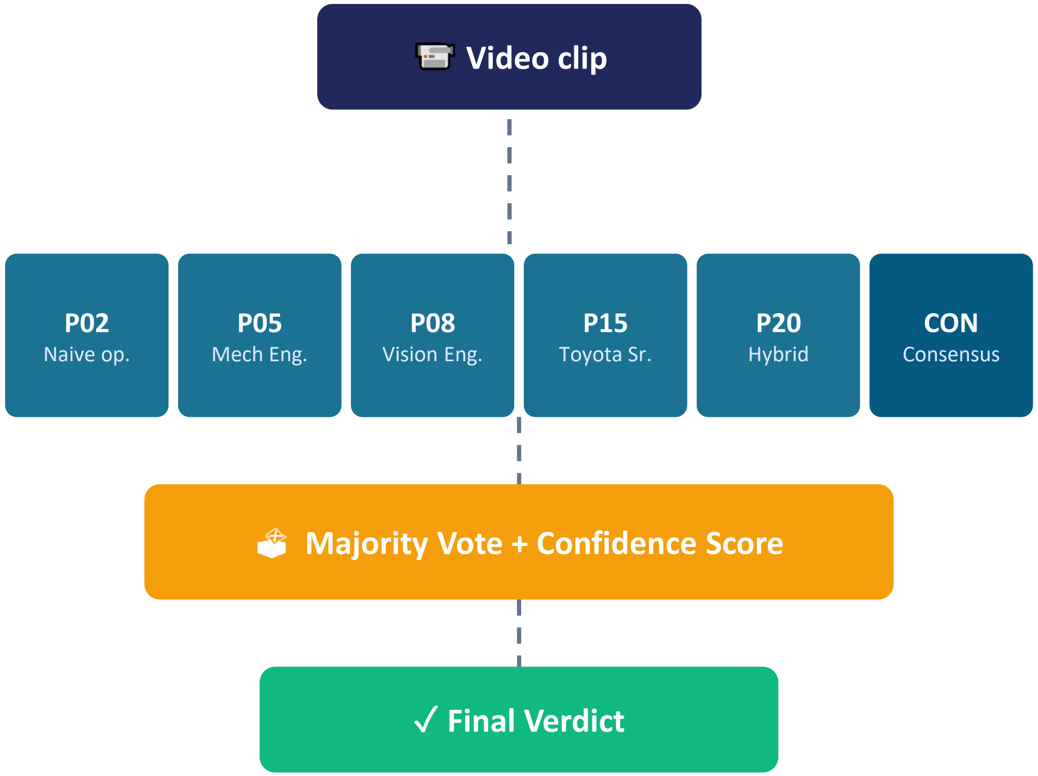
```
SHARED_CHECKLIST = """
COMPONENT GEOMETRY (read this first - it defines how to interpret the scene)
==============================================================
The push-button is a SMALL CYLINDRICAL component with two distinct ends:
  * RED CAP END  - one circular end is a flat RED DISC. This is the part
                   a human would press to actuate the switch. Visually
                   it appears as a SOLID RED region on one side of the
                   button.
  * PIN END      - the other circular end is a BLACK plastic face with
                   4 small metal pins (the electrical contacts).
                   Visually it appears as a DARK region with small
                   pin protrusions visible.

THE BUTTON RESTS ON THE BELT IN A SIDEWAYS POSITION - this is the
EXPECTED carriage state for this part. Do NOT interpret "lying on its
side" as a tilt, fall, or structural failure. This is how the button is
SUPPOSED to travel down the belt.
```

```
"""
20 different quality control inspector personas, each describing the same
push-button conveyor belt scenario from their own professional perspective.

All personas share the same anomaly rules:
  - Correct: RED CAP on the RIGHT, PIN SIDE on the LEFT
  - Anomaly: pins on the right (reversed), stopped, fallen, slid off the belt

But each uses different vocabulary, different emphasis, and different focus.
This provides "synthetic persona diversity" for model ensembling.

Usage:
    from personas import PERSONAS, VOTING_SUBSET, CONSENSUS_PROMPT

    for persona in VOTING_SUBSET:
        prompt = persona["prompt"]
"""
```

Complete Persona Prompts

Single-prompt VLM responses are unreliable on edge cases. We adopt the self-consistency / persona-prompting paradigm:

1. **Synthesize 20 inspector personas**
   line operator, mech engineer, vision engineer, Six Sigma QE, lean specialist, conservative auditor, …
2. **Select 5 maximally-diverse voters**
   plus a 6th consensus prompt distilled from all 20
3. **Run inference per voter, parse verdict, vote**
   majority decides; tie or mixed = LOW confidence flag



Three Prompt Designs

Specialized

Each voter sees one aspect

PROS

- Deep per-aspect reasoning
- Lower prompt overhead

CONS

- Systematic blind spots
- Voters can miss anomalies outside their domain

Complete

Every voter checks A / B / C

PROS

- No blind spots
- Multi-axis verification

CONS

- Verbose prompts
- Slower inference
- Compact YES/NO bias

Orientation-Only

Single binary question per voter

PROS

- Sharpest signal
- Fastest inference
- Best alignment with dominant defect

CONS

- Cannot detect motion / position anomalies

Future Work

Custom Conveyor Belt Dataset Build a labeled benchmark using a DIY setup (cardboard + pens + fixed camera) covering three anomaly classes: Missing Cap, Wrong Orientation, and Stuck Conveyor. Enables quantitative evaluation through precision, recall, and F1 metrics.

Hybrid Approach (VLM + Classical CV) Combine VLM's semantic reasoning with OpenCV-based detection (color, edge, motion). Classical CV handles geometric defects deterministically; VLM focuses on contextual reasoning. Faster, cheaper, more reliable.

Monitoring Dashboard Web-based UI displaying verdicts, confidence scores, and per-persona votes. Includes anomaly logs and historical defect statistics for line operators — turning the system into a usable product.

References

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- Qwen Team, "Qwen3-Omni-30B-A3B-Instruct" [online model], Hugging Face, https://huggingface.co/Qwen/Qwen3-Omni-30B-A3B-Instruct, 2025.
- Wei et al., "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models", NeurIPS 2022 (arXiv:2201.11903)